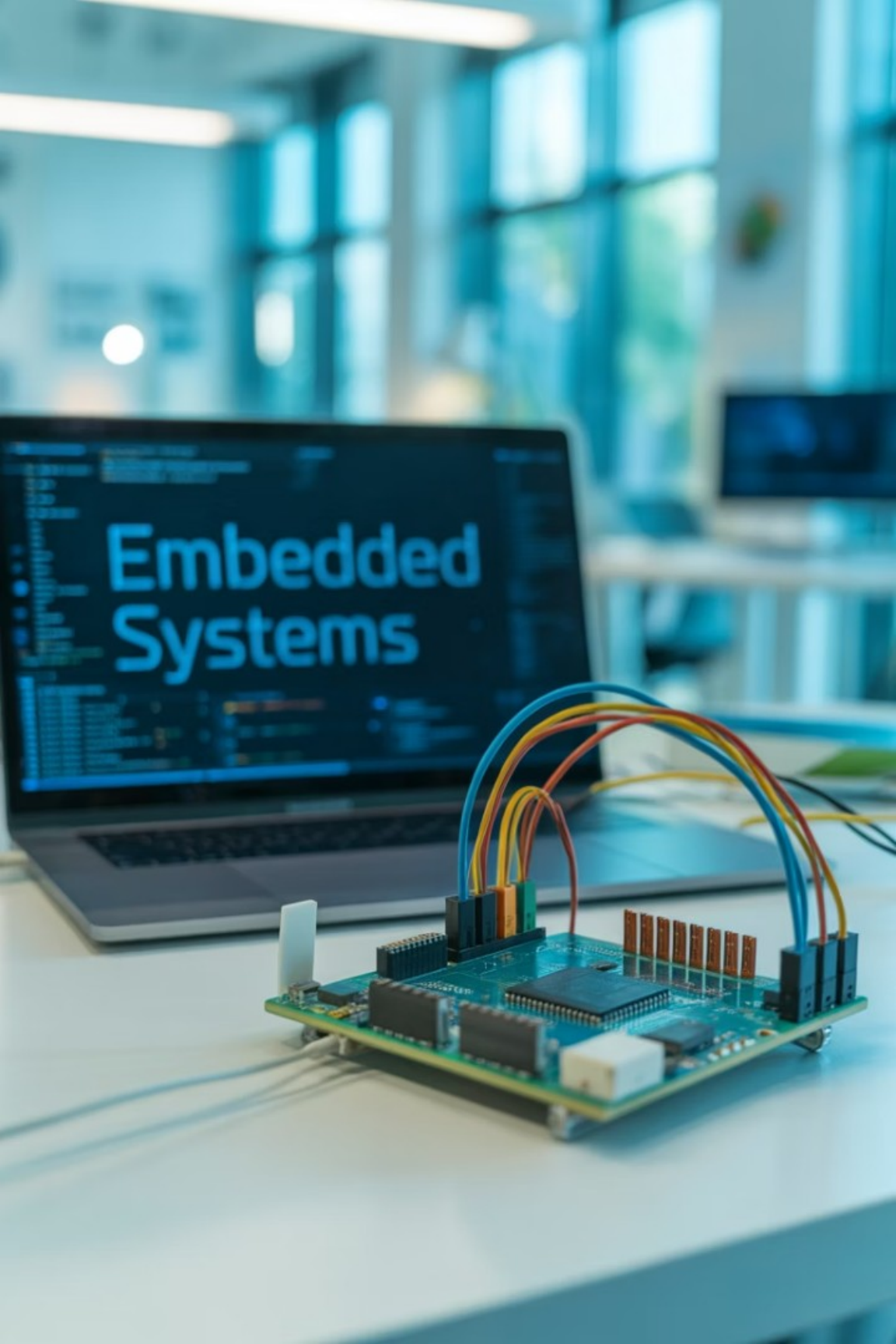




Zephyr RTOS Intro





Achieve Your Goals

Agenda

1

What is Zephyr RTOS?

Small, scalable open-source RTOS

2

Key Features

Architecture, boards, connectivity

3

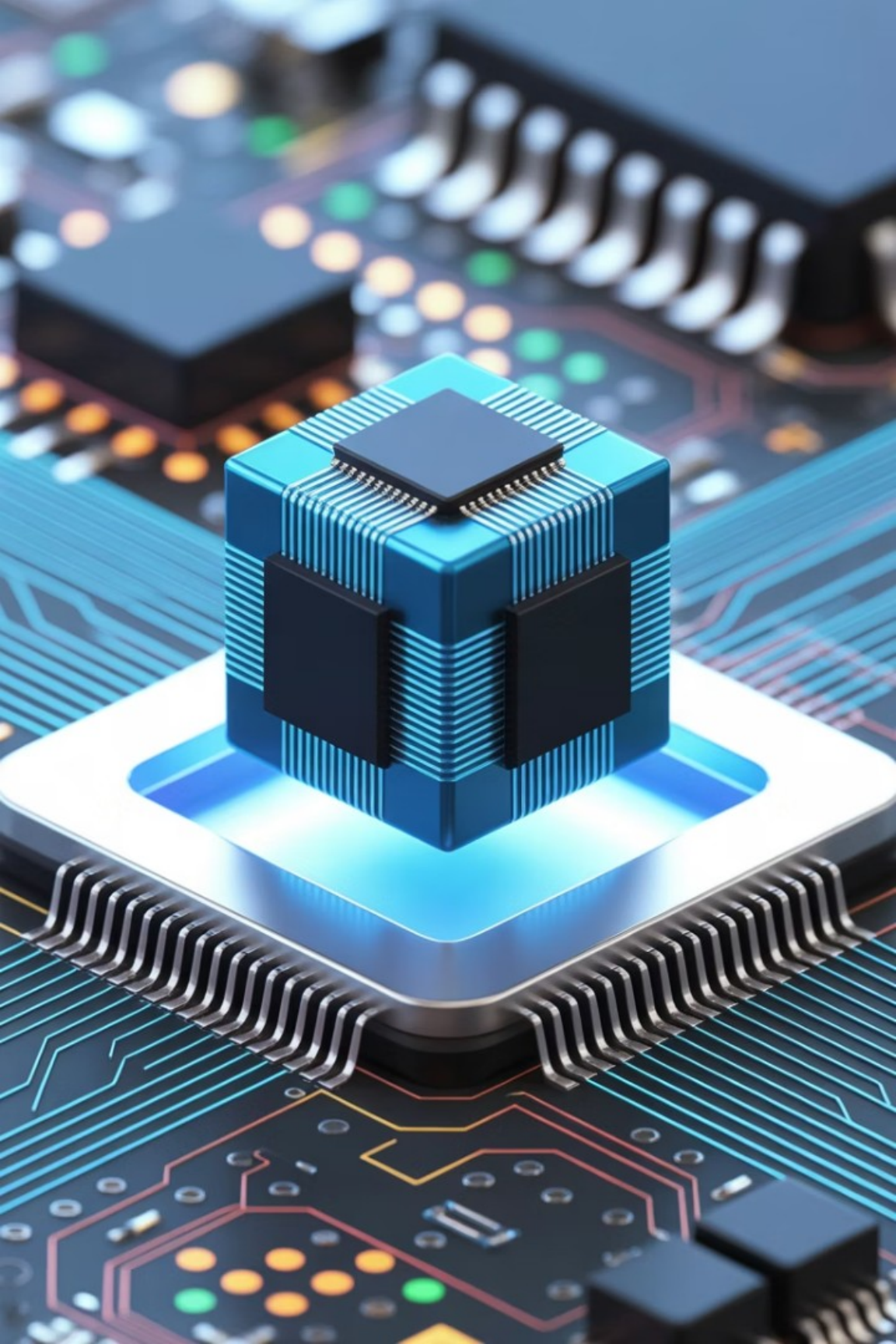
Advantages

Compared to alternatives

4

Real-World Use

Products, trends, getting started



What is Zephyr RTOS?

Small & Scalable

Designed for resource-constrained embedded systems

Open Source

Vendor-neutral, community-driven

Wide Support

ARM, RISC-V, x86, Xtensa

Heritage

Originated from Wind River's Rocket OS (2015)

Key Features

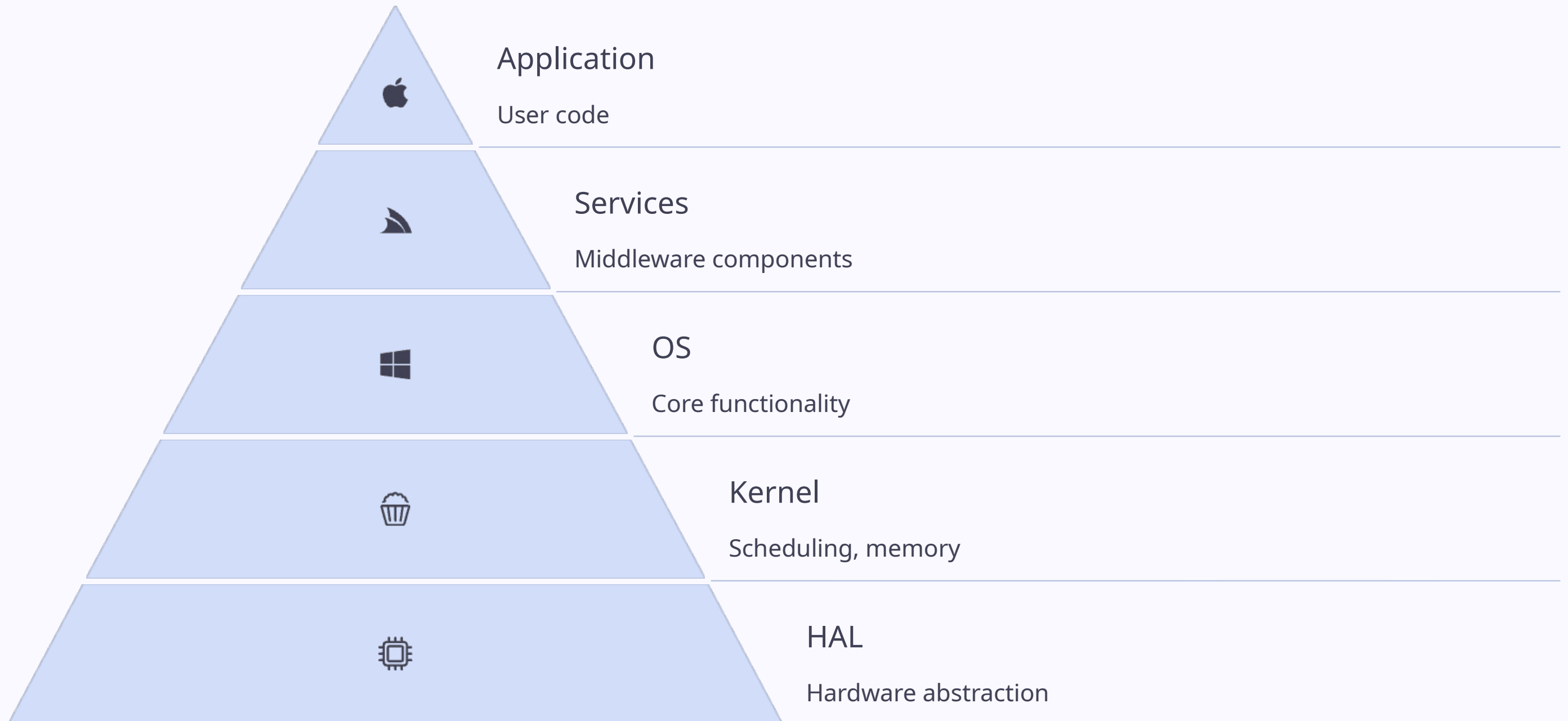
Minimal Footprint

- <8KB Flash, <5KB RAM
- Scalable to multi-core
- BLE 5.3, Wi-Fi, CAN, USB

Extensive Support

- 750+ boards, 220+ sensors
- CMake, Kconfig, Devicetree
- Full protocol support
- Power management

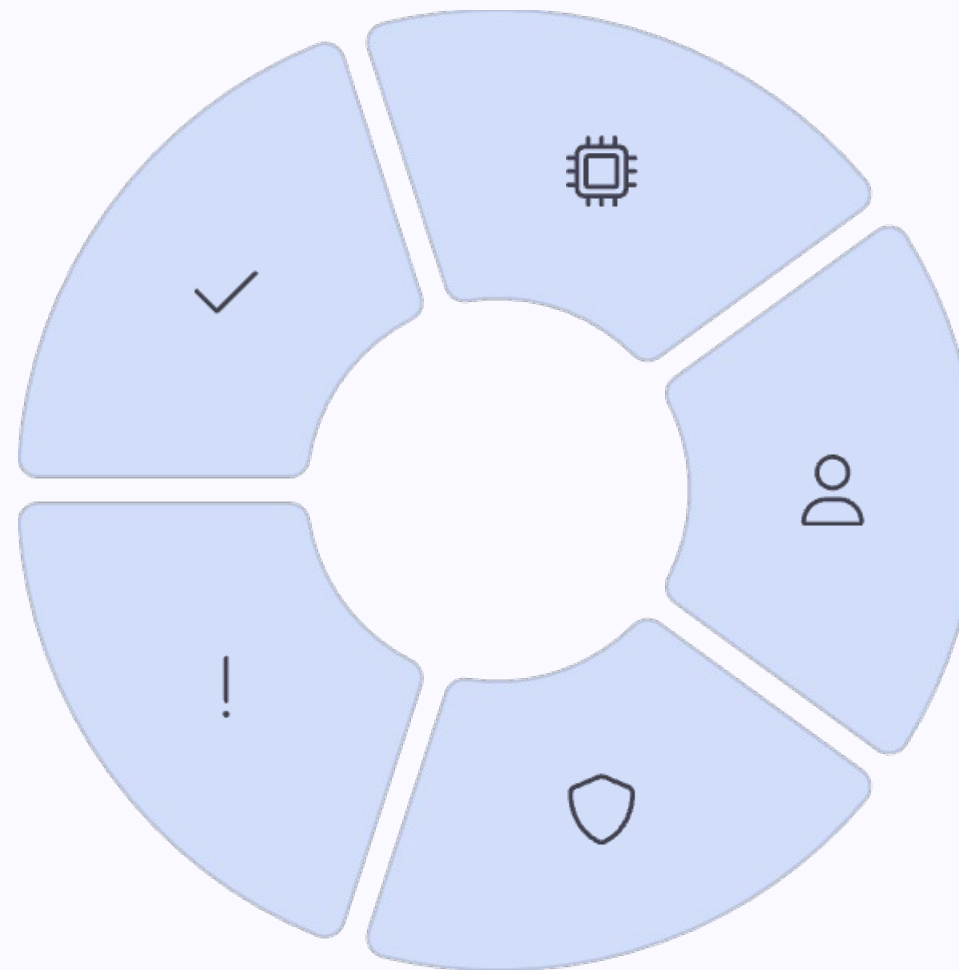
Architecture Overview



Zephyr vs Other RTOSes

Open Source
Apache 2.0 vs commercial licensing

Challenges
Steeper learning curve, newer ecosystem



Hardware Support

750+ boards vs 40-100 for others

Community

1,100+ contributors, 5x growth since 2017

Security

Dedicated PSIRT and vulnerability management



Advantages over Alternatives

Feature	Zephyr RTOS	Linux	Bare Metal
Memory	KBs	MBs	Minimal
Boot Time	Milliseconds	Seconds	Immediate
Determinism	High	Low	Manual
Portability	High	Medium	None
Power	Efficient	Hungry	Variable

Real-World Products Using Zephyr



Consumer

Galaxy Ring, Moto Watch 100



Computing

Chromebook EC, Framework Laptop



Industrial

Vestas, BLiXT, IoT devices

25+ Million Devices in the Field



Getting Started



Setup

Install deps, setup west, get SDK



Develop

Clone repo, choose sample, target board



Deploy

Build/flash/debug with west



Learn More

Docs, GitHub, Discord